HW2 Bonus

Wednesday, February 11, 2026 7:25 PM

4.8.4

a) Given : Predict test obs' response using only 10% of range of X closest to that test obs & X is uniformly dist. ← $P(a \le X \le b) = b-a$

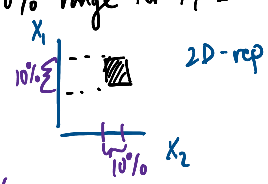
∴ empirically we are using 10% of the data to make a predict'n like in the ex. case w/ the range extending 0.05 from both directions of $X = 0.6 \rightarrow [0.55, 0.65]$

~ $[X_0 - 0.05, X_0 + 0.05]$; however what about boundary cases like $X = 0.01$ resulting in $[0, 0.06]$ where range is shortened...

★ The key lies in the Q asking for the 'avg.' behavior, which focuses on expected interval length. Therefore avg. across $[0,1]$, we use 10% of available obs. to make a pred.

b) given $X_1 = 0.6 \rightarrow [X_1 - 0.05, X + 0.05] = [0.55, 0.65]$
$X_2 = 0.35 \rightarrow$ " " $= [0.3, 0.4]$ } $p = 2$ features

Excluding boundary cases, an obs. must be in the 10% range for X_1 & 10% range for X_2 to be considered for prediction.



Therefore, as an obs. must satisfy BOTH cond. & assuming X_1 & X_2 are indep. we can represent the % of observations used to make predictions is $0.10 \times 0.10 =$ 1%

c) following the logic from a & b. We can state the avg. % of obs. used for prediction given $p = \#$ features is $0.10^p = \frac{1}{10^p}$

Therefore given $p = 100$, the fraction of available obs. we use for pred. is $\frac{1}{10^{100}}$ or 1e-100

d) if we consider r% of observations around a feature as 'near' to be considered for obs. & p as the # of features, our formula for available % of obs. that can be used to predict any given test obs. is : r^p

- since $r \in [0,1]$; r likely takes a smaller val. on the range since the value represents 'nearness'
- raising r to a large p drives $r^p \rightarrow \emptyset$; as seen by how $0.1^{p=100}$ allows us to use functionally 0% of the training obs. for a given test obs.
- the smaller r is, the faster the 'decay' towards $\emptyset$ is.
- However, regardless of the value of r (given $r < 1$), an inc. in p causes the fraction of 'nearby' obs. to shrink exponentially towards $\emptyset$. (i.e. the available train obs. w/in a fixed 'radius')

~ KNN relies on finding 'close' observations to the test point. (from training obs.). In higher dimensions, what can be considered 'close' rep. a much smaller % of the training data. In other words, the K nearest neighbors may actually be quite far away which makes them poor predictors for test data. Essentially 'nearness' doesn't offer as much value as p - # of dimensions increases.

e) ~ the problem essentially asks the inverse of a) - c)

- p-dim. hypercube w/ side length ℓ has volume $\ell^p \overset{set}{=} 10\%$ observations

~ solving for ℓ : $(\ell^p)^{1/p} = (0.1)^{1/p} \rightarrow \ell = \frac{1}{10^{1/p}} = \sqrt[p]{1/10}$

- at $p = 1$, $\ell = \frac{1}{10} = 0.1$; need 10% of range for hypercube that contains on avg. 10% train obs
- at $p = 2$, $\ell = \frac{1}{10^{1/2}} = \sqrt[2]{1/10} \approx 0.316$; need 31.6% "
- at $p = 100$, $\ell = \frac{1}{10^{1/100}} = \sqrt[100]{1/10} \approx 0.977$; need 97.7% "

These findings tell us that in order to capture 10% in higher dimensions, the hypercube needs to extend across an increasing proportion of the traing obs. (i.e. the feature space).
In this case, the data used to predict a traing obs. is no longer actually 'near' or 'local' as p becomes large which defeats the purpose of KNN.